Growth and Development of The Infant and Toddler

Infancy Stage

Third Stage
Dr. Moustafa Alshammari
Growth and Developments



Assessment of Normal Growth & Development of The Infant

- A number of methods have been proposed to assist the nurse to evaluate that growth progresses in a regular and orderly sequence.
- Tables, charts and grids are available to facilitate the nurse's evaluation of physical growth.
- In addition to the observation over a period of time, the periodic assessment of infants and children permits the early detection of growth deficiencies and pathologic processes, both organic and psychological.

[1]. Growth During The Intrauterine Period

The period of intrauterine life may be divided into two principle phases:

- **The embryonic period**: this is considered to be the **first 8 weeks** during which the ovum is fertilized and beginning of the period of organogenesis.
- **The fetal period: 8th to 40th or 42nd week** of gestation is distinguished by rapid growth and elaboration of function.

[2]. Growth During The Postnatal Period. (Newborn Assessment)

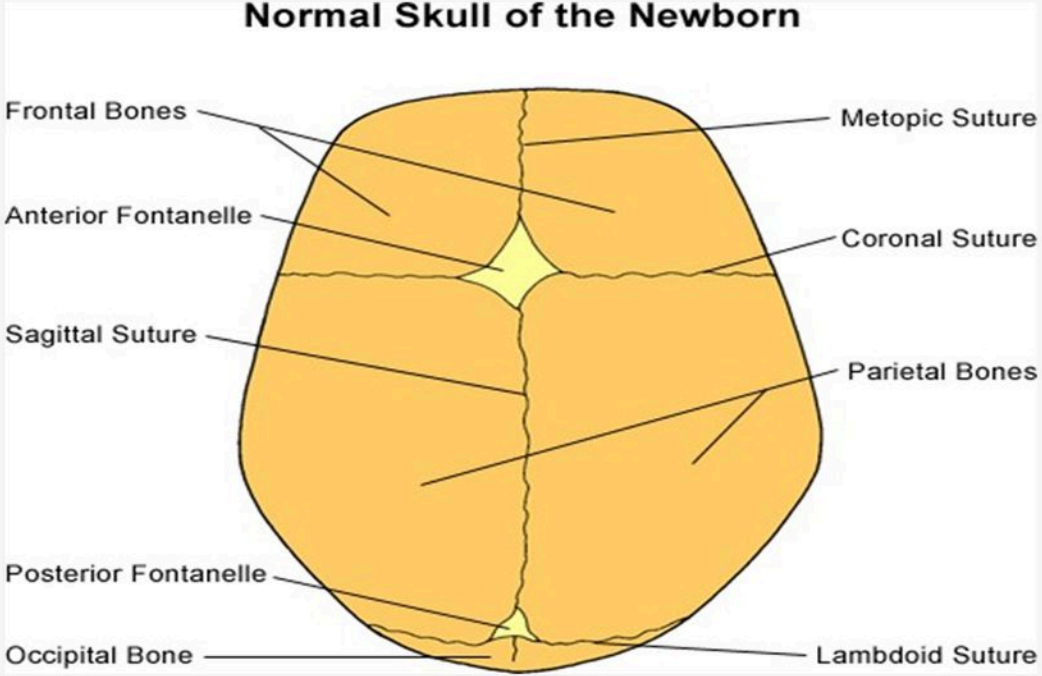
1. Growth of head:

- The posterior fontanel closes at 1-2 months, Triangular shape
- The anterior fontanel closes at 12-18 months.
- At birth: 3.5 - 4 cm. Diamond shape.
- At birth, head circumference is 33-35 cm.
- At the end of the first year, it is 45 cm.
- At the second year, it is 48 cm.
- At the third year, it is 49 cm.
- At the fifth year, it is 50 cm. At the seventh year, it is 52 cm. At twelve years, it measures 53 cm.
- And the adult head circumference is 55 cm.

Head circumference measurement



Main fontanels on the head of the newborn



Growth of Chest and Abdomen:

2. The chest and abdomen circumferences are equal in the first one to two years of life.

- The abdomen tends to be prominent in infants and young children.

3. Chest Circumference:

- At birth, head circumference is 2cm larger than chest circumference.
- Between age of 1 and 2 years both are equal.
- After 2 years, the chest is larger than the head.
- N.B: Chest is usually measured at nipple line, with the subject sitting in the midway between inspiration and expiration.

4. Growth in length:

- During the first year in life, the infant gains in length roughly 2 cm/month during the first 6 months and 1.5 cm in the second 6 months.
- ***At birth: 48-50 cm**

- A formula can be used after the first year to estimate the length of the child **(Age in years x 5)+80=length in centimeters**.

5. GROWTH IN WEIGHT:

- There is an initial phase of loss of weight after birth for the following factors:
- 1. Withdrawal of hormones from mother.
- 2. Loss of excessive extra cellular fluid.
- 3. Passage of meconium (faeces) and urine.
- 4. Limited food intake.
- Most full terms regain their birth weight by the age of ten days. Then the weight gain averages approximately 20 – 25 gm/day.
- A formula can be used after that to calculate the weight:
  (Age in years X 2) + 8 = Weight in kg.

B. PHYSIOLOGICAL GROWTH:

Vital Signs:

- a - Temperature: will drop immediately after birth in adjustment to delivery room temperature unless the newborn infant is kept warm immediately after birth.

  Temperature rises to normal within about 8 hours. Temperature of newborn infant is usually 35.5 to 37.5 °c.
- b - Pulse: 120 to 150 / min. It is usually rapid and irregular.
- c - Respiration: 35 to 50 / min. It is usually irregular in depth, rate and rhythm.
- d - Cry with no tears.

DEVELOPMENT:

- Bodily functions and responses to external stimuli are carried out by reflexes.
- ** Reflexes: The reflexes such as: Moro-reflex, swallowing and gagging reflex. Sucking reflex, grasp reflex , tonic-neck reflexes.

Emotional Development:

- Newborn infant expresses his emotion just through cry for hunger, pain or discomfort sensation.

Social Development:

- Cry is his contact with environment; to communicate his physiological needs as hunger, throaty sounds show interest in human face.

Infant Normal Growth & Development

- Assessment of normal growth.
- Development is assessed as:
  - Gross motor.
  - Fine motor (sensory motor).
  - Personal-social.
  - Language.
  - Cognitive.

NORMAL INFANT

- Physical growth :

a - Weight gain:

- Birth to 4 months - 3/4 kg / month.
- 5 to 8 months - 1/2 kg / month.
- 9 to 12 months - 1 /4 kg /month
- He will double birth weight by 5-6 months and triple it by 10-12months of age.

Normal Infant

length

- **During the first year in life, the infant gains in length roughly 2.5cm cm/month during the first 6 months of age. Average height is 65 cm at 6 months and 75cm at 12 months**
- **Increase by about 25 cm in 1st year**

Growth of Head:

- **The posterior fontanel closes at 1-2 months.**
- **The anterior fontanel closes at 12- 18 months.**
- **Cranial sutures**
  - **The sutures between the cranial bones do not ossify until later childhood**
- **Head circumference increases about 1.5 cm/month during the first 6 month of age, then 1/2cm /month during the second 6 months of age.**
- **It is *43* cm at 6 months. At the end of the first year, it is *46* cm.**

CHEST CIRCUMFERENCE:

- **At the end of the first year, chest circumference and head circumference are equal .**
- **N.B: Chest is usually measured at nipple line, for greater accuracy, take two measurements- one during inspiration and the other during expiration-and record the average .**

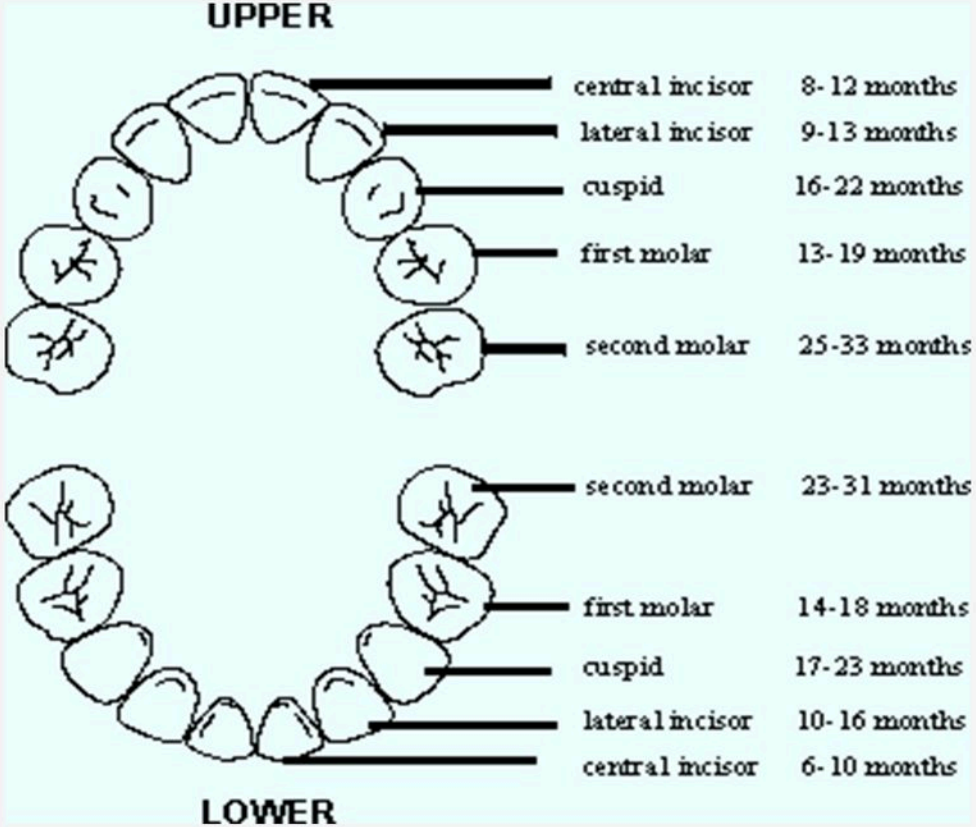
Eruption of The Milk Teeth (Deciduous Teeth)

The development of human dentition is a continuous process from 5th month in utero to maturity

By 5–7 months of life, The first deciduous teeth usually erupt

- Babies may differ in the timing of tooth eruption
- Teething is a normal continuous process of development and does not cause fever or respiratory problems

Eruption of The Milk Teeth



Gross Motor Development

1 month Raises head momentarily when prone, hips extended

2 months When prone raises chin off couch

3 months In a prone position, he will rest on forearm, keep head in midline, make crawling movements with legs, arches back, and supports head when held erect.

4 months Sits with support. Moro reflex disappears.



Cont. Gross Motor Development.



6 months Puts feet in mouth in supine position. Starts crawling.

7 months Sits without support leaning forward



8 months Rolls over from prone to supine. Sits well with no support

9-10 Creeps and pulls self upright.

11 months Stands unsupported.

12 months Walks supported. Toddles' way and cruises around furniture.

Walks well.

Starts crawling upstairs.

MOTOR DEVELOPMENT IN INFANTS



9-11 months

creeping



12 months

walks

supported



Parachute

reflex

6-12 months



Walks well

12 months

Fine Motor Development

1 months Follows a moving object to midline. Responds to sounds by blinking.

2 months Hands often open. Grasp reflex is fading

3 months Hands mostly open. Regards his hands. Can hold a rattle if placed in his hand. Follows 180° by eye.

4 months inspects and plays with hands. Tries to reach objects with hands. Grasps objects with both hands. Can carry objects to mouth

5 months Turns head to sound. Brings hands together.

6 months Able to reach object by hand and get it. Grasps feet and pull to mouth

Cont. Fine Motor Development

7. months Can transfer a rattle from hands to hand. Reaches objects and brings them to mouth. Has ambidextrous approach

8 months has beginning pincer grasp. Can feed himself with a biscuit.

9 months Tries picking things with fingers.
Preference for use of dominant hand

10 months Crude release of an object beginning.
Grasps bell by handle.

11 months Can use a thumb and finger to pick a cube. Grasps by thumb and finger.

12 months Can pick small objects “pellets "No longer puts objects in mouth. Builds a tower of 2 blocks but fails. Throw objects. Holds cup to drink.

Psychosocial Development of The Infant

- First develops a sense of trust when fed on demand
- By 2 months distinguishes mother or primary care giver
- Eventually learns that not every need is met immediately on demand

- By 3 to 4 months: smiles in response to smile of others .shows interest in other family members
- By 7 to 8 months :shows fear of strangers (strangers anxiety).
- By 9 to 10 months: play simple games with adults ,e. g. ‘’bye-bye’’. play with adult games such as ‘’peek-a-boo’’ by 10 months

’peek-a-boo” by 10 months



- He learns to trust others through the relief of basic needs, i.e., to trust those who give pleasant sensations.

- Sense of trust will result also from being held, talked to, cuddled, warmed, and so... on.

- If this sense of trust in others is not learned, the reverse, a sense of mistrust is. acquired

Health Promotion of The Infant

- Nutrition requirements
- Promoting sleep
- Dental care
- Accident prevention
- Toy selection

Supplemental Nutrients That May Be Prescribed For The Infant

- Vitamins C and D
- Iron
  - Breast-fed infants need supplements of iron, as well as vitamin D
  - By 6 months of age, iron-rich foods are needed as supplements
- Fluoride

Components of Health Promotion and Maintenance of The Infant

- Routine checkups
- Immunizations
- Family teaching
- Education about accident prevention

Well-baby Visits

- Occur at 2 weeks, and at 2, 4, 6, 9, 10 and 12 months
- The nurse collects data regarding:
  - Growth and development (weight, height, head circumference)
  - Nutrition and sleep
  - The caregiver–infant relationship
  - Any potential problems
- Immunizations are given to guard against disease
- Family teaching is provided

Health Protection of The Infant

Immunization

- **BCG (Tuberculosis)**
- **Hepatitis A and B**
- **Polio**
- **DPT (Diphtheria, Pertussis, and Tetanus**
- **Measles**
- **German measles (rubella)**
- **Mumps**
- **Varicella (chickenpox)**
- **Haemophilus influenza meningitis (HIB)**
- **Pneumococcal disease**

Injury Prevention

- Falling down
- Aspiration
- Poisoning
- Suffocation
- Burns
- Motor vehicle accidents

Health Promotion of The Infant

a. Toy selection

- **Infant toys serve several purposes, including:**
- **a) Stimulation for psychosocial development**
- **b) Diversion for relieving boredom, pain, and discomfort**
- **c) A means of communicating and expressing feelings**
- **d) Aid in the development of sensor motor skills**

Dental Health

- 1. An infant's primary (deciduous) teeth erupt at about age 6 months. Assessment guide: age of child in months minus 6 months equals' number of primary teeth.
- 2. Clean an infant's teeth with a damp cloth; brushing is too harsh for the infant's tender gums.

- **3.** Assess the need for a fluoride supplement; consult with the physician.
- 4. Despite a widespread belief to the contrary, fever, vomiting, and diarrhea usually are not associated with teething but rather indicate illness.

Thank you

رمز الصف

cs5jm5o